MILITARY SPECIFICATION SHEET

ELECTRON TUBE, PHOTOTUBE

TYPE 922

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Red-infrared sensitive, vacuum

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ebb	Light source	Light	Rp	T	Ik	Average cathode current density (Jk)
Unit:	Vdc	---	ℓm	Meg	°C	μAdc	μA/sq in.
Maximum:	500 (Note 7)	---	---	---	100	5 (Note 5)	30 (Note 6)
TEST CONDITIONS:	250	LS1	0.1 (Note 1)	1.0	---	---	---

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
5303	Spectral response identification	---	S1	---	---	---	---
1331	Direct-interelectrode capacitance	---		Cpk	---	1.5	pF
1031	Low-frequency (25 Hz) vibration	---	No voltages	---	---	---	---
	Quality conformance inspection, part 1						
5301	Anode current	2		Ib	1.2	4.0	μAdc
---	Gas ratio	2,3		GR	---	1.5	---
5301	Dark current	2	Light = 0	LIb	---	0.005	μAdc
	Quality conformance inspection, part 2						
(C) 1105	Permanence of marking	---		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	4	Group C: Ebb = 500 Vdc; t = 500 hours (min)	---	---	---	---
---	Life-test end point:						
5301	Anode current	---		Ib	0.8	---	μAdc

NOTES:

1. The light during test, except in the case of life-test operating conditions, shall be incident upon a 0.5-inch diameter aperture located at the center of the cathode.
2. This test to be performed at the conclusion of the holding period.
3. Gas ratio = (Ib at Ebb = 250 Vdc)/(Ib at Ebb = 25 Vdc).
4. The luminous flux at the cathode shall be adjusted to give an anode current of 4.0 microamperes on the average tube. Color temperature shall not be adjusted to less than 2,000°K.
5. Average cathode current. Averaged over any interval of 30 seconds maximum.
6. Average cathode current density. Averaged over any interval of 30 seconds maximum.
7. An ac supply voltage may be used having a peak value equal to this maximum value.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

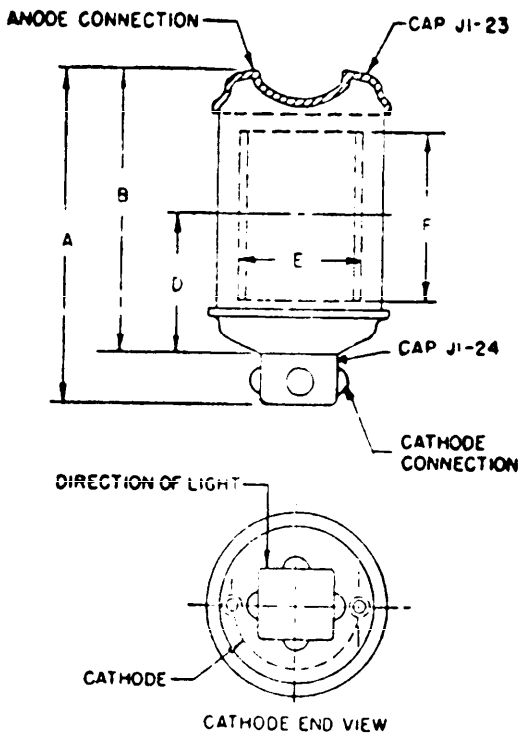
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2557)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
B		1.438(36.53)
E	.50 (12.70)	
F	.625(15.88)	
Quality conformance inspection, part 2		
A		1.719(43.66)
D	.624(15.85)	.750(19.05)

FIGURE 1. Outline drawing of electron tube type 922.

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DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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DEPARTMENT OF THE NAVY
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-E-1/410C

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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